



Bio-inspired Adaptive Locomotion via Torque-based Learning

A Case Study of SATA

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2026 / 05 / 25

Command



Torque



Motion

1 Locomotion Is a Control Problem

THE PROBLEM

Position-based locomotion is accurate under known conditions but rigid — it struggles with compliance, disturbances, unknown terrain, and the sim-to-real gap.

RIGID CONTROL ON UNKNOWN TERRAIN



→ Torque control may improve compliance, providing conditions for more adaptive locomotion.

POSITION CONTROL

Command position

Rigid behavior

Poor adaptation

TORQUE CONTROL

Command force

Compliant interaction

Adaptive response

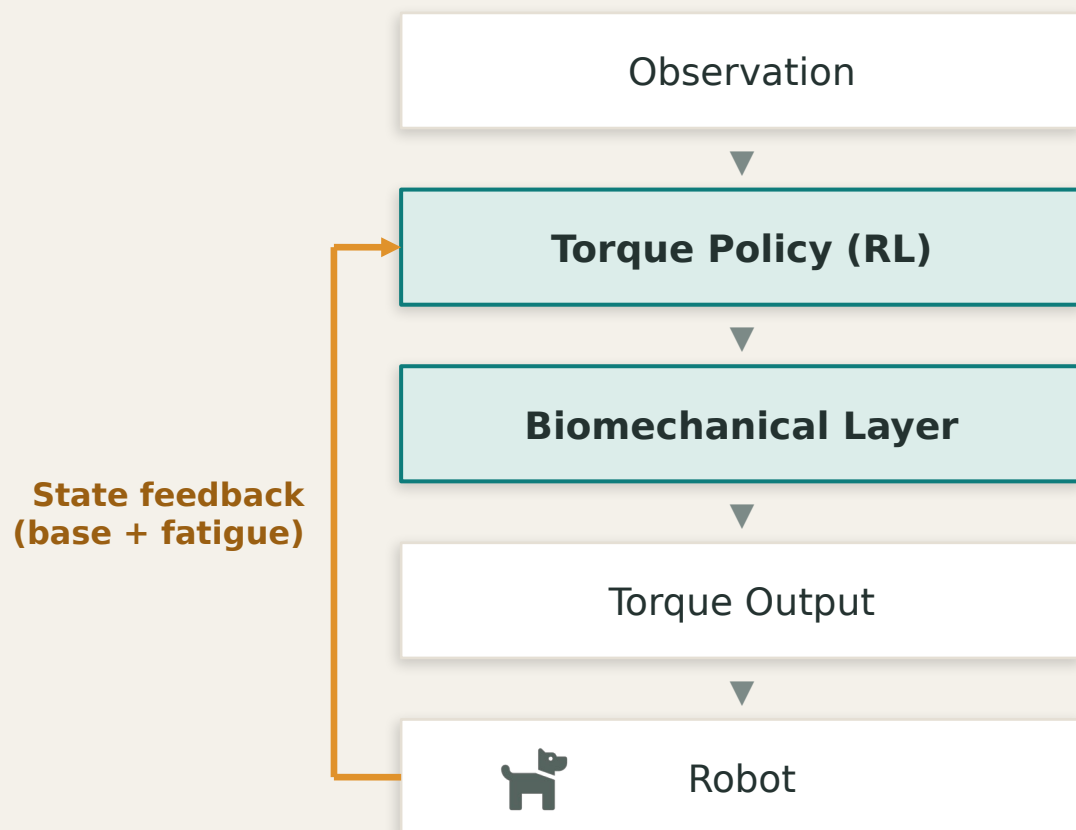
Adaptive behavior $\neq$ adaptive control

RESEARCH QUESTION

How can robots achieve safer and more adaptive locomotion in unknown environments?

2 SATA: Bio-inspired Adaptation in Torque Control

An RL torque policy wrapped by a biomechanical layer and a growth curriculum.



Biomechanical Model

 activation

 muscle

 fatigue



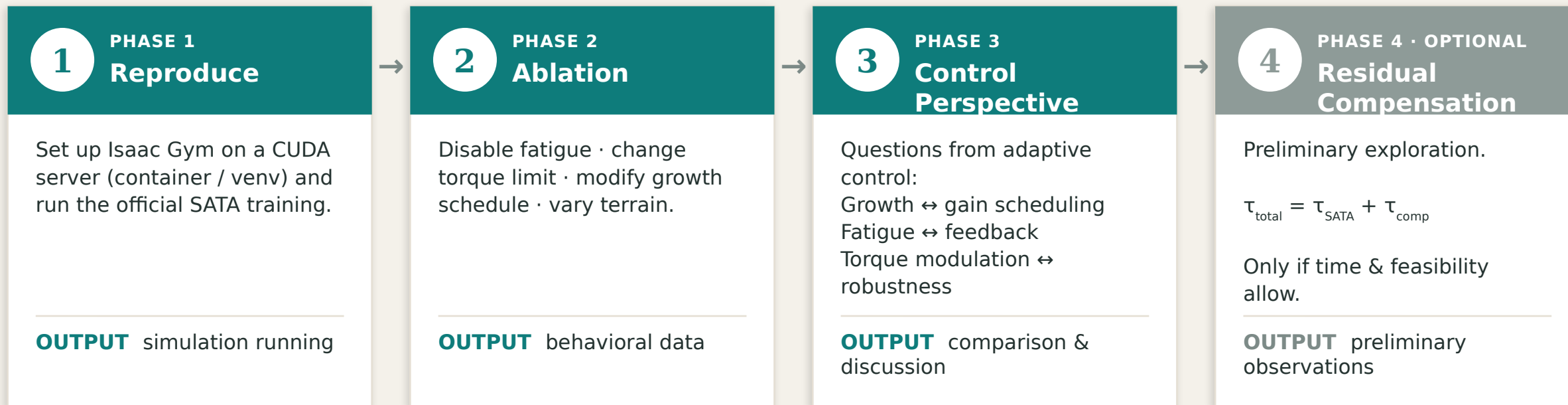
Growth Mechanism

torque limit · training curriculum · control frequency

Goal: smoother torque generation and improved robustness through bio-inspired adaptation.

3 Plan: Understand → Reproduce → Analyze

A feasible 3-phase study, with an optional extension — we study SATA, not redesign the RL.



APPROACH

We analyze adaptation instead of proposing new methods — we do not redesign the RL algorithm.

4 Expected Outputs & Contribution

We study adaptive behavior — not propose a new RL algorithm.

EXPECTED OUTPUTS

- 1 Simulation reproduction
- 2 Behavior analysis (ablation)
- 3 Control-perspective comparison
- 4 Literature-grounded discussion

RESEARCH QUESTIONS

RQ1

Why torque control?

RQ2

What creates adaptation?

RQ3

How does SATA address problems traditionally studied in adaptive control?

RQ4

Can lightweight compensation help?

IN ONE LINE

This project studies how bio-inspired torque control produces adaptive locomotion, and interprets it through adaptive and robust control perspectives.